

5G and 5G evolution

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LA FORZA DELLE CONNESSIONI

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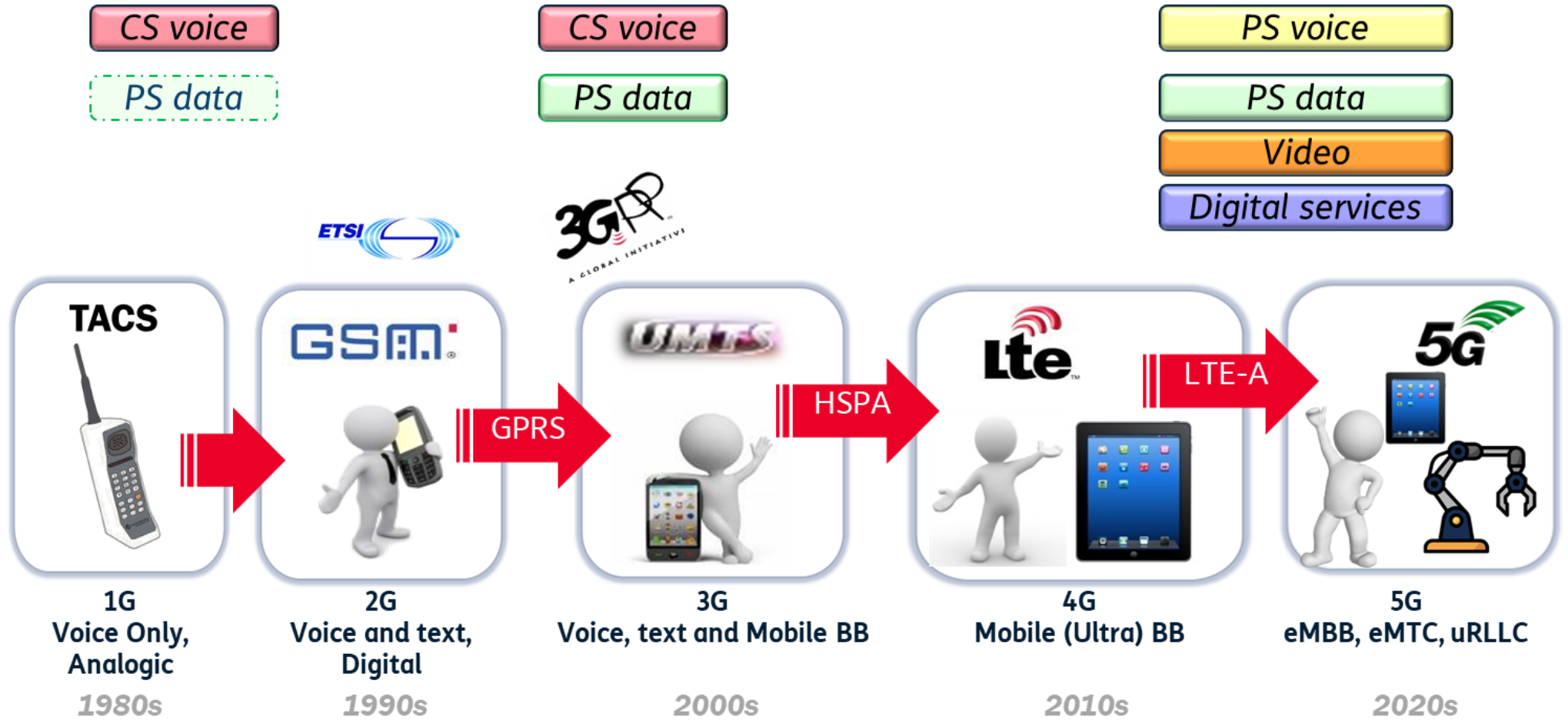
5G Evolution towards 6G



5G Usage Scenarios and KPI



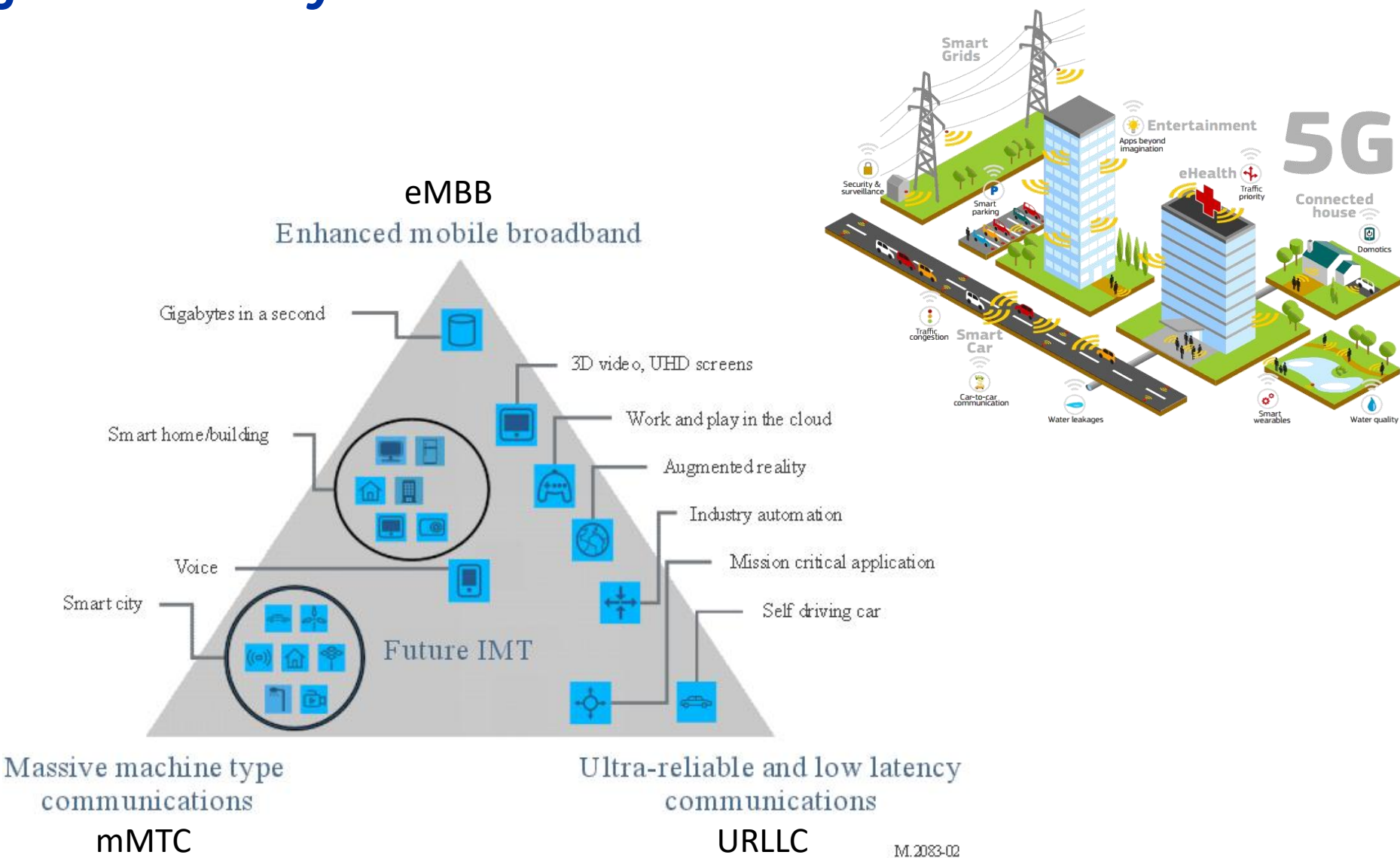
Mobile Cellular System Evolution Path



PS : Packet Switched
CS : Circuit Switched
BB: Broad Band

eMBB : enhanced Mobile Broadband
eMTC : enhanced Machine Type Communication
uRLLC: ultra Reliable and Low Latency Communications

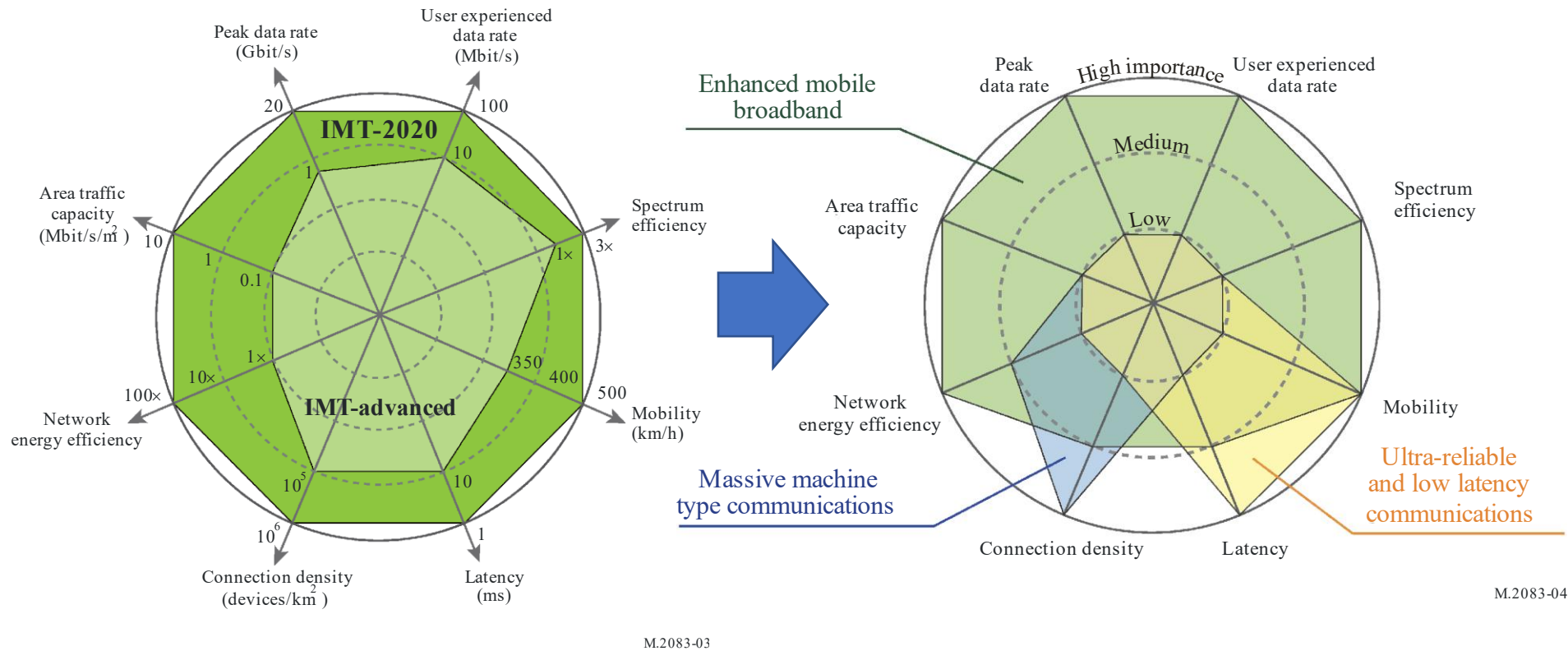
5G: Usage scenario beyond Mobile Broadband



Source: Racc. ITU-R M.2083, "IMT Vision", 2015

Expected KPI for 5G

Not only improved performance (or «doing better» what we can do with LTE)



Source: Racc. ITU-R M.2083, "IMT Vision", 2015

Mapping KPIs to Usage Scenarios



enhanced Mobile Broadband (eMBB)

- 10-20 Gbps peak
- 100 Mbps user experienced data rate
- 10000x more traffic
- Macro and small cells
- Support for high mobility (500 km/h)
- Network energy saving by 100 times



Massive Machine Type Communication (mMTC)

- High density of devices (2×10^5 – 10^6 /km²)
- Long range (MCL 164 dB)
- Low data rate (1 – 100 kbps)
- M2M ultra low cost
- 10 years battery life

Enabled by NB-IoT and eMTC



Ultra Reliable and Low Latency (URLLC)

- Ultra responsive
 - < 1 ms air interface latency
 - 5 ms E2E latency
- Ultra reliable and available (99.999%)
- Low to medium data rates (50 kbps – 10 Mbps)
- High speed mobility

Mapping KPIs to Usage Scenarios



enhanced Mobile Broadband
(eMBB)



Massive Machine Type
Communication (mMTC)



Ultra Reliable and Low Latency
(URLLC)

Smart Office

Virtual Reality
+
Augmented Reality

Ultra HD 8K

Smart Cities

Smart Lighting
Smart Parking
Smart Wasting

Smart Metering

Smart Logistic

Tactile Internet

Smart Health

Autonomous Driving
+
Industry 4.0

Remote Controlling

Public Safety



3GPP 5G Roadmap



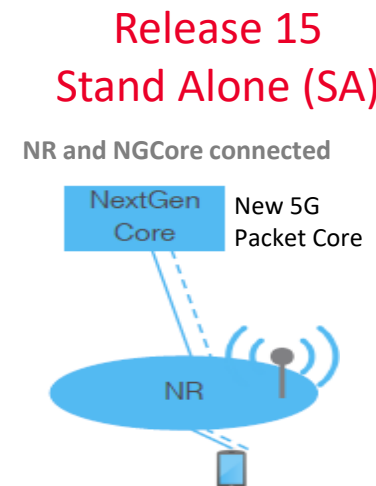
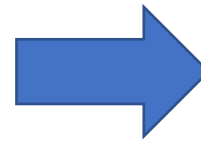
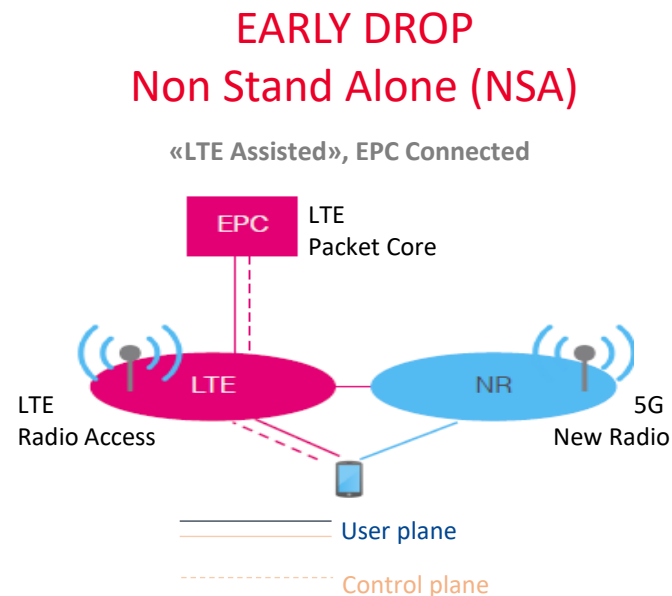
3GPP 5G Roadmap timing rationale



- Rel-15 (i.e. **5G Phase 1**) was the first release of 5G specifications and was overall completed in **December 2018**
- It mainly targeted **enhanced Mobile Broadband (eMBB)** and basic **ultra Reliable Low Latency Communications (uRLLC)**
- It followed a **step-wise approach** based on **drops** which allowed for all the **5G architecture options** to be specified
- The Rel-15 **Early Drop** (December 2017) focused on the Non Standalone (NSA) NR architecture
- The Rel-15 **Regular Drop** (June 2018) included support for NR Standalone (SA)
- The Rel-15 **Late Drop** (March 2019) completed the specification of the remaining 5GC-connected architectures
- **Release 16** (aka phase 2, by June 2020) targeted the ITU IMT-2020 submission

The Early Drop

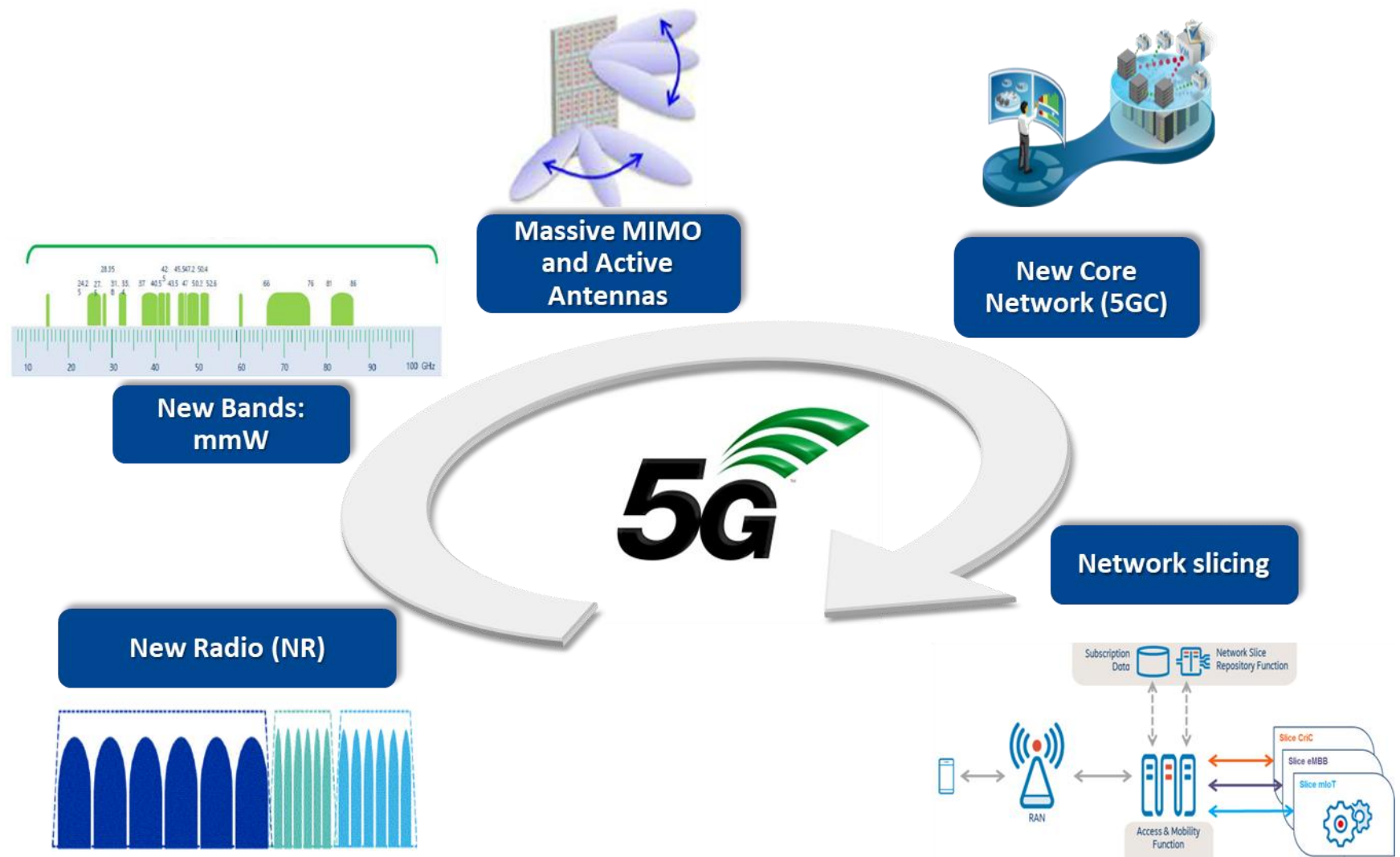
- Early phase 1 (Early drop) specified in Dec '17: mainly eMBB
- Initial standard solution for 5G New Radio (NR) addresses non-standalone “hotspots” and small cells where (e)LTE plays the role of the primary communication technology providing wide area coverage and control plane
 - Using LTE core network
 - LTE eNB always acts as a master, NR gNB always acts as a slave
- NR provides radio capacity boost to reach higher data rate in hotspots



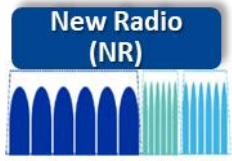
Key Technological Enablers



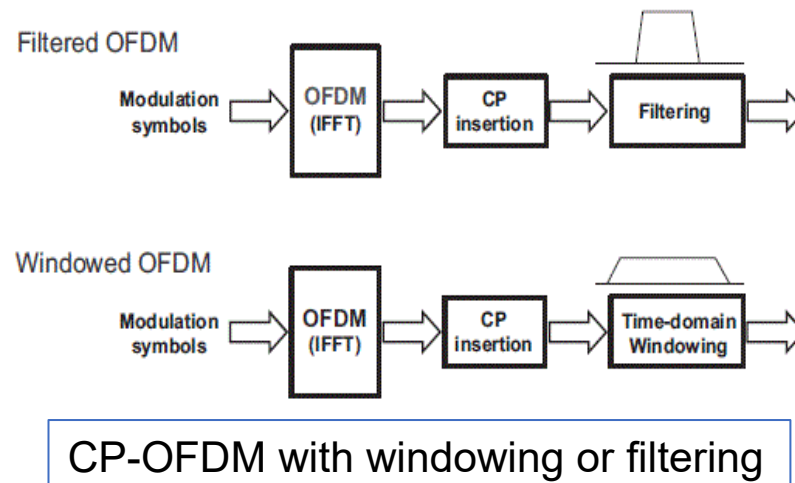
5G key technological enablers



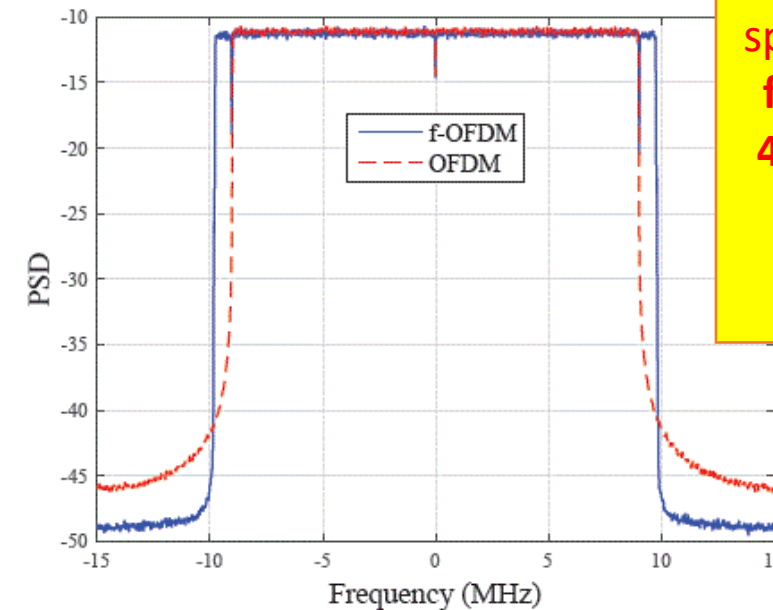
New Radio (NR): 5G filtered OFDM



- 3GPP has agreed to adopt **OFDM** with a **scalable numerology** (subcarrier spacing, cyclic prefix) in both UL and DL **up to at least 52.6 GHz**
- **Windowing or filtering** operations that are transparent to the receiver are applied **on top of CP-OFDM** at the transmitter side to **improve spectrum confinement** (i.e. **filtered OFDM**)



Ex: 20 MHz channel



More efficient spectrum use:
from 90% in 4G to 98% in 5G !



New Bands for 5G communications

- Current specifications define two different frequency ranges in which NR can operate:

Frequency range designation	Corresponding frequency range
FR1	450 MHz – 6000 MHz
FR2	24250 MHz – 52600 MHz

Auction for spectrum in Italy in september 2018.

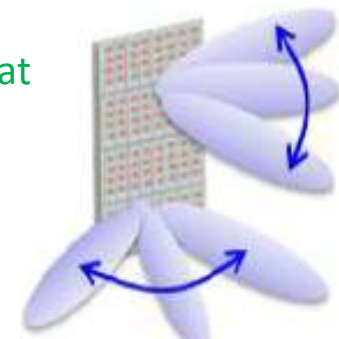
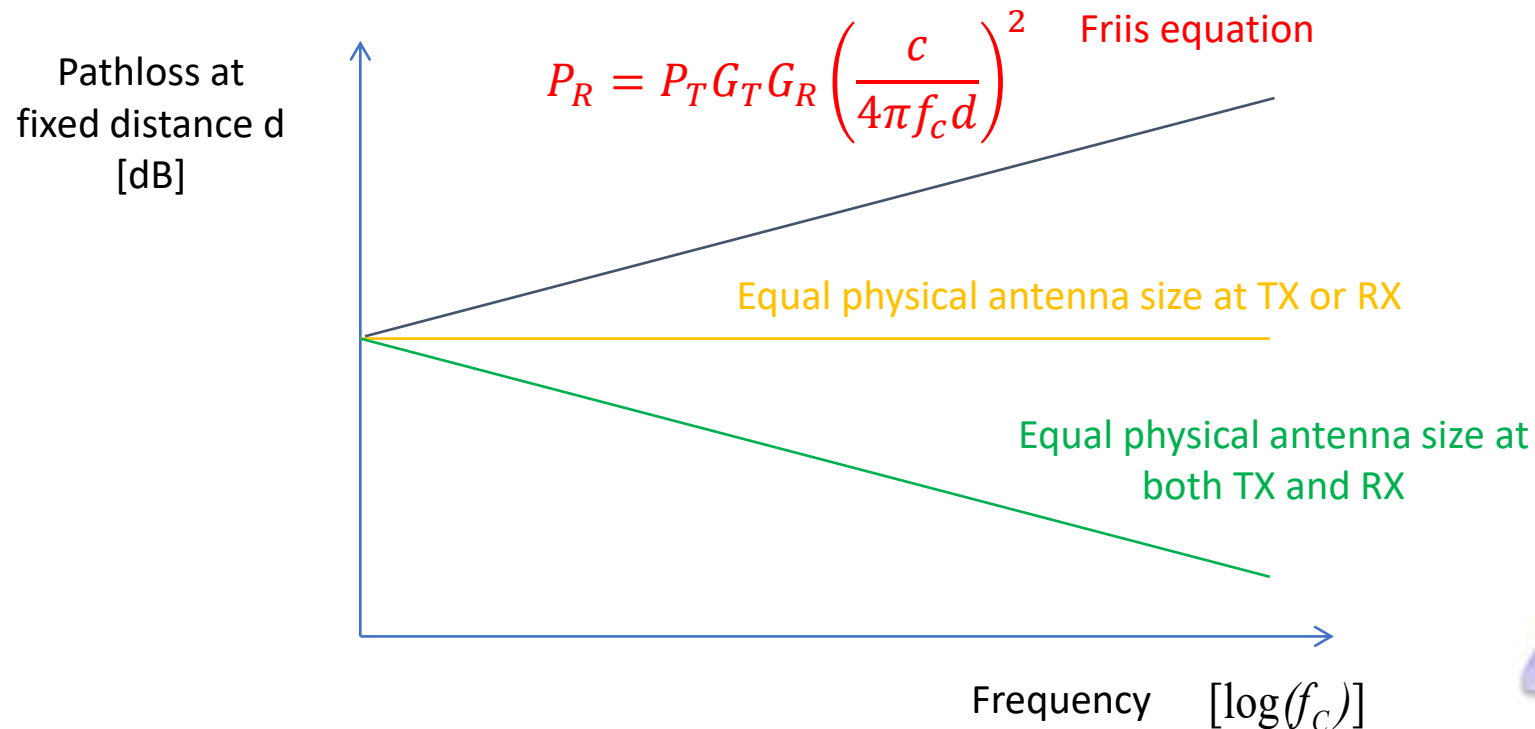
5G Frequency Auction in Italy (6.55 bln €)

	694-790 MHz	3.6-3.8 GHz	26 GHz
Frequency and duration	▶ 30+30 MHz FDD* ▶ 1/07/22 - 31/12/37 (15.5 years)	▶ 200 MHz TDD ▶ 1/01/19-31/12/37 (19 years)	▶ 1 GHz TDD ▶ 1/01/19-31/12/37 (19 years)
Lots and assignees	▶ 2x10 MHz: reserved for new entrants – assigned to Iliad ▶ 2x10 MHz assigned to TIM; 2x10 MHz assigned to Vodafone	▶ 80 MHz TDD assigned to TIM; 80 MHz TDD assigned to Vodafone ▶ 20 MHz TDD assigned to Iliad; 20 MHz TDD assigned to Wind3	▶ 200 MHz TDD: one lot assigned each to TIM, Vodafone, Wind3, Iliad, Fastweb

mmWave challenges: Pathloss



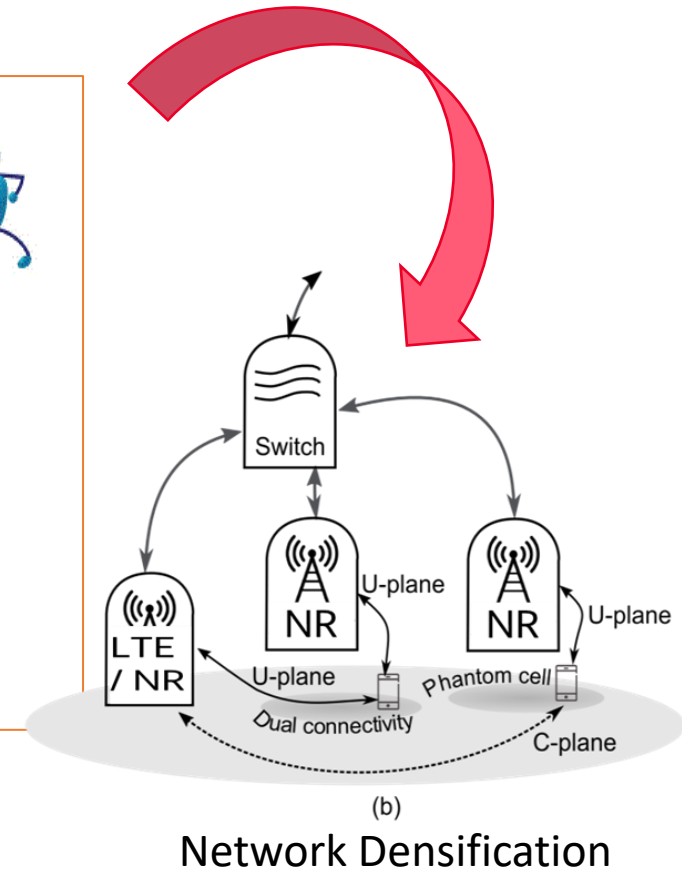
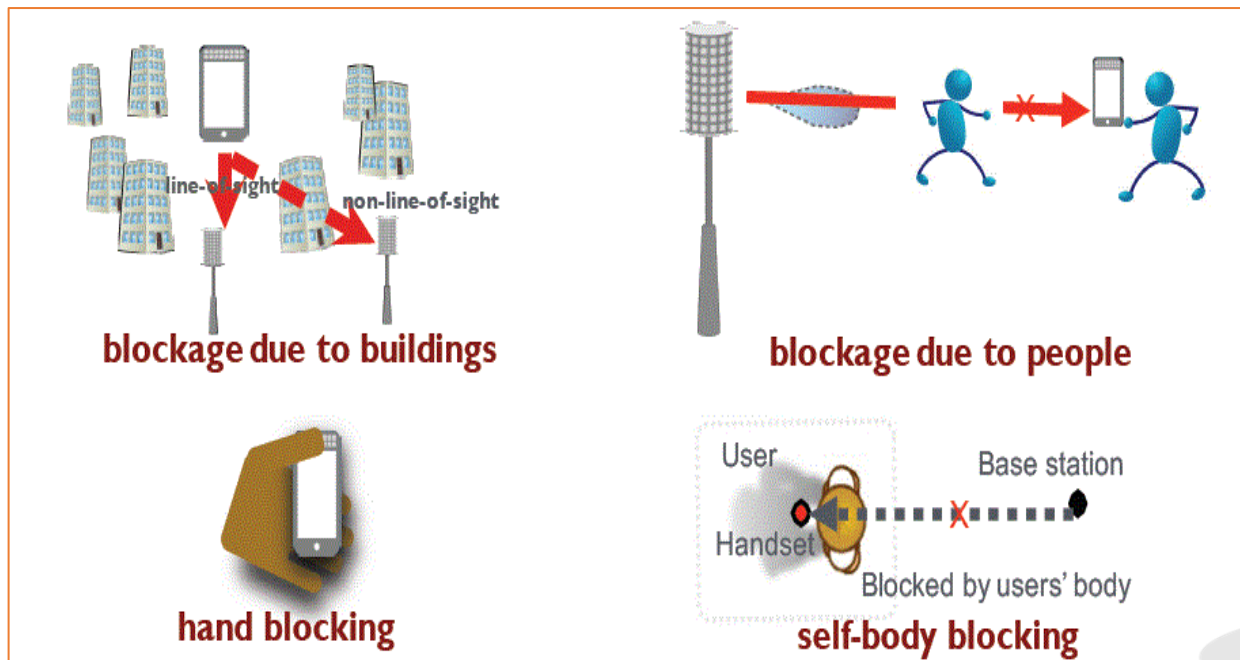
- **Increasing** the carrier frequency **increases the power loss** for a given distance d
- However **Maintaining the same physical size of the antennas** (by increasing the number of antennas) allows to **keep constant the pathloss** or reduce it as the frequency increases



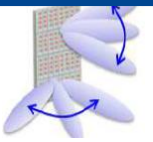


mmWave challenges: Blockage

- **Electromagnetic waves** have also **weak ability to diffract** around obstacles
- With a small wavelength, **mmWaves are sensitive to blockage** by obstacles (e.g. buildings, humans, etc.)

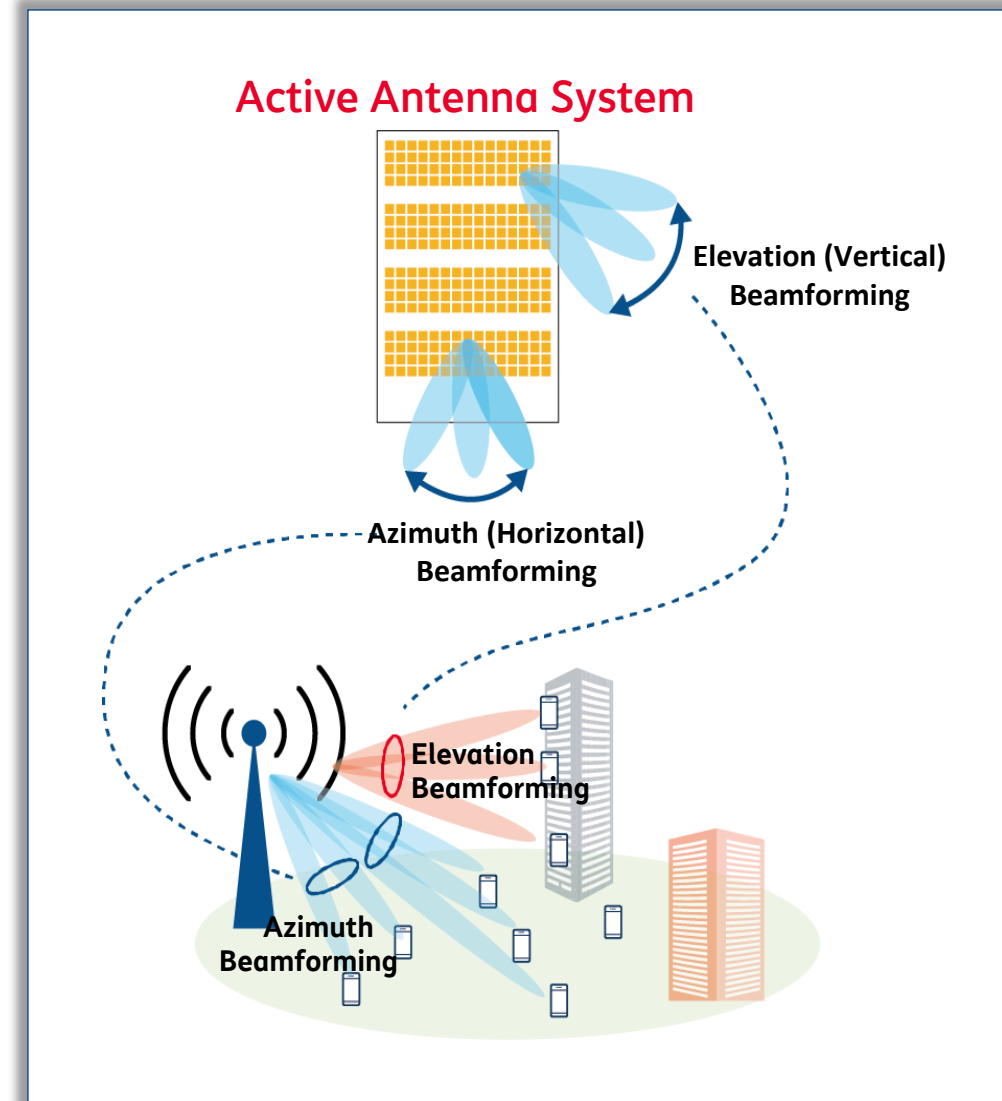


Source: "Millimeter Wave for 5G Features and implications".
Professor Robert W. Heath Jr.

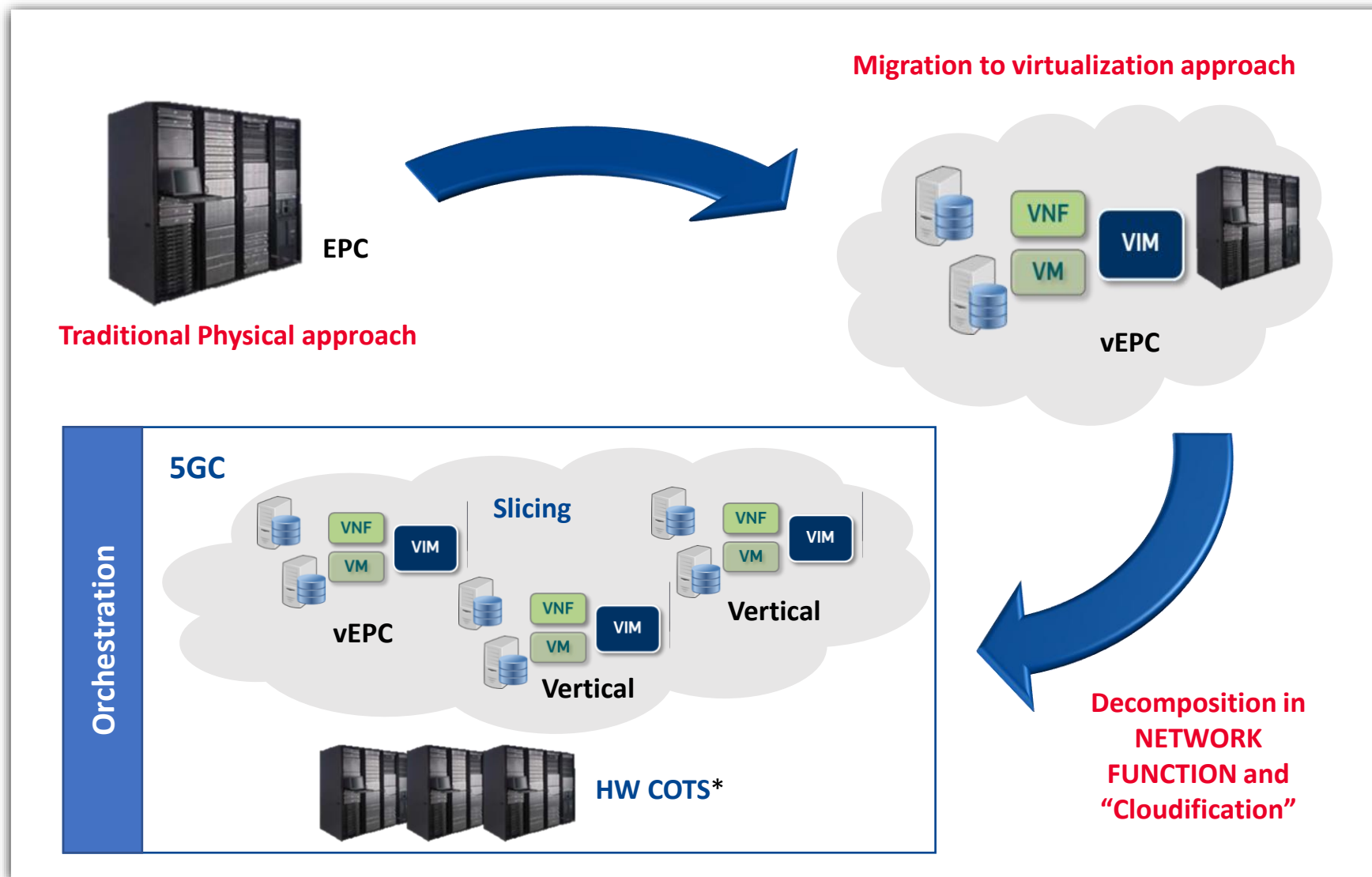


Massive MIMO and Active Antennas

- **Active Antenna System**
 - array antennas with «active» elements. Each radiating element connected to a radio transmission chain
- **Beamforming**
 - tens, hundred, thousand of «active» elements allow beamforming capability
- **New M-MIMO antennas are a complex pillar of NR technology**
- **M-MIMO antennas bring also operational challenges:**
 - Cost
 - Installation challenges, e.g. due to weight
 - Power Consumption
 - EMF human exposure



New 5G Core Network – “cloudification” process



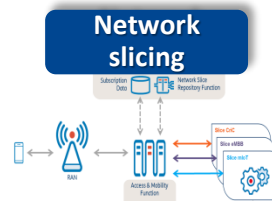
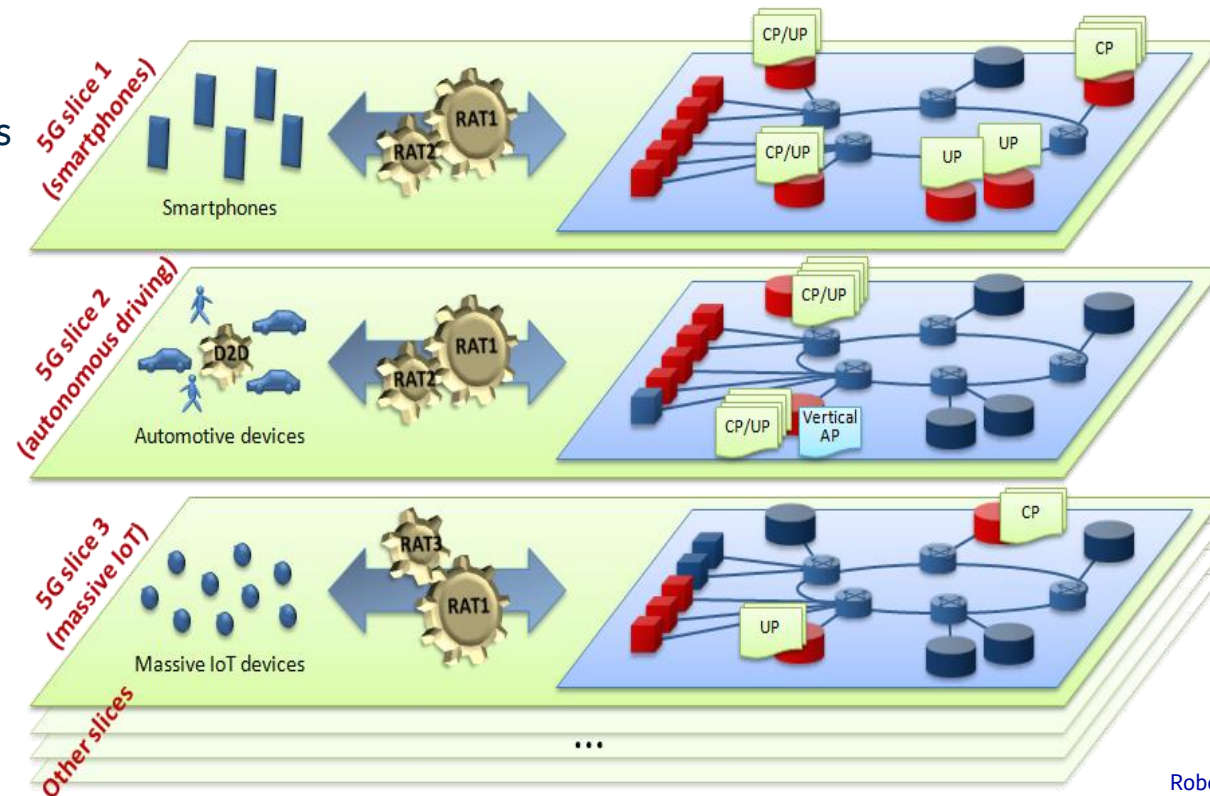
*COTS = Commercial Off-The-Shelf

Network Slicing

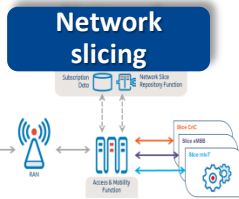
- A **5G network** architecture with **native support for NFV** enables **fast creation of dedicated network** instances **optimized for specific customers, service** scenarios and/or **terminal** types (**network slices**), potentially improving user experience and infrastructure resources
- **Network slices** could be created by the operator **combining elementary functions** available **in a catalogue. Composition of functions** to create slices can be managed by the operator **though the NFV Orchestration Platform**
- Integration of multiple RATs in the network slices, including WiFi

Examples of slices:

- ▶ **Smartphone:** fully-fledged functions distributed across the network
- ▶ **Automotive:** security, reliability, latency at the edge. Cloud node + open API for vertical applications.
- ▶ **Sensors:** avoid mobility, basic Control plane functions (e.g. authentication)

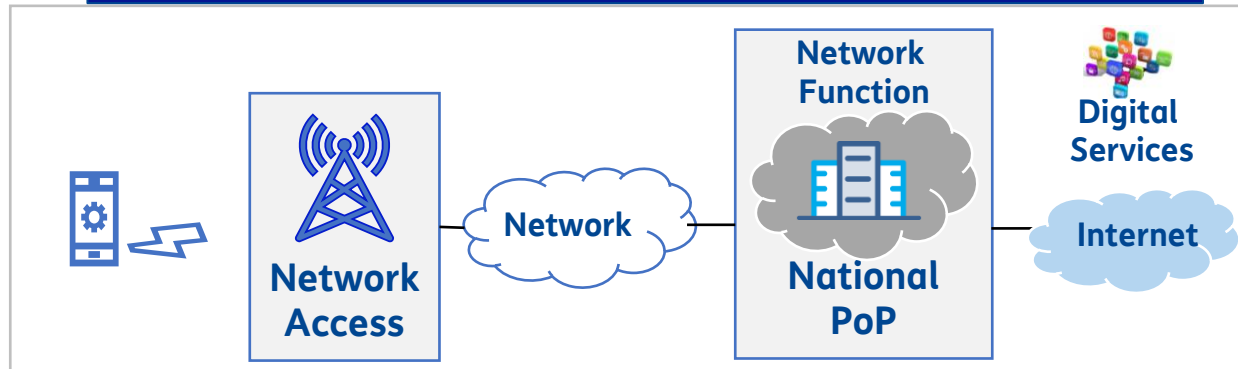


Edge Computing – Architecture definition and related Benefits

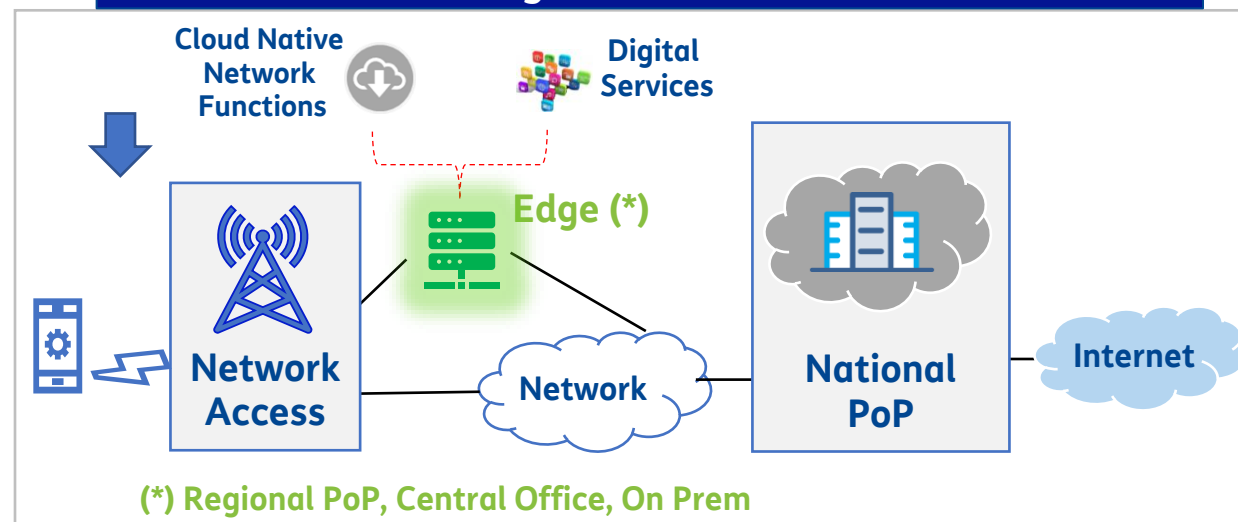


Edge allows to **run services close to the end user**, improve application response time, security and performance

Centralized Architecture



Edge Architecture

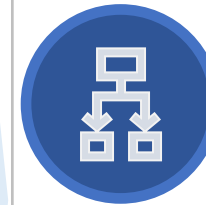


(*) Regional PoP, Central Office, On Prem

Edge Computing Benefits



New Services: Enabling new digital services thanks to **performance, proximity and security**



Local Data processing: **Reliability, security** and **segregation of data** enable a secure perimeter for services (such as video surveillance, Smart Cities, Private Networks...)



Computational balancing: Deploying Services break out at the edge enables **efficient balancing** of the computational load **between device and network**



Performance / Latency Optimization: Moving platforms closer to customers **improves CEX** and **reduces user-network interaction time**

5G use case examples



UseCase – Virtual reality– AR/VR Maintenance

Scenario

(on-field assistance with remote support from an expert)

When carrying out maintenance or inspection of an apparatus, the operator - wearing an AR headset and with free hands – can be assisted with overlaid information about the operations that needs to be performed.

Furthermore, the operator can request the assistance of more experienced personnel, who can remotely share what the operator sees in order to give assistance both vocally or by interacting with the 3D reconstructed scene

Applicable to all standardized operations:

- Allows intervention by non-expert personnel;
- Can reduce travel, training and intervention costs;
- Can produce intervention certification by recording each step



UseCase – Virtual Reality – Immersive Virtual Museum Visit

Scenario

The visitor, wearing an appropriate Virtual Reality headset, will be able to visit some historical / tourist places made available by Museums / Public Administrations (e.g. the Sassi of Matera)

Within the experience, the visitor will be free to move around in the planned areas, approach the objects manipulation tables, or move between the different points of interest.

The experience can take place in two ways:

- Experience assisted by a Virtual Guide, a sort of “narrative voice” that guides the visitor illustrating a predefined visit route
- Interactive experience, meeting in the virtual world a real museum guide (availability to be agreed) which makes the visit much more engaging, allowing the conversation in real time between the guide and the visitor, as well as the exchange of virtual objects



UseCase – Virtual Reality – Video 360° live

Scenario

Allow the fruition of events (e.g. sports competitions, concerts,...), creating an immersive experience made possible through virtual reality headsets.

Through the use of 360 cameras, which allow shooting in High Definition of 360 ° live video, video flows are generated that can be enjoyed in immersive mode with virtual reality headsets, both in real time and on-demand (*)

The video stream is sent over the network to a 360 video management and publishing platform. The broad band and the stability of the 5G network connection guarantee a smooth and continuous transmission, allowing the simultaneous viewing of numerous visitors.

Visitors can connect at any time and from anywhere, finding themselves immersed in an event / place as if they were there, and moving freely can look around. 360 videos can also be enjoyed in non-immersive mode, via PC browser and smartphone.

(*) in addition to the 360 cameras, additional cameras can be used such as action-cams, wearables that send "only" HD streams, but which can be worn or mounted on moving objects (cars, bikes, drones, ...) to expand the virtual experience of the event. People who participate to the event (or spectators) could send their contributions in real time to the content publishing platform via smartphone (and then be selected / published by the control room to the various Social).



UseCase – Public Safety – BodyCam Wearable

Scenario

The UseCase is contextualized for use in the Public Safety field. Agents are equipped with a wearable BodyCam capable of recording images related to actions in the field (events, interventions, ...). The images can:

- Be acquired and downloaded in a second moment when returning to the operational headquarters (e.g. police headquarters) where, through a specific SW and a suitable docking station, they can be cataloged (also as evidence in case of Process) and archived. This mode has already been made available with 4G devices (LTE)
- Be acquired and streamed live to a control room in the operating office (eg police headquarters). The sending is at present made possible under wi/fi coverage or 4G coverage (LTE).

5G coverage (and related devices) further enhances the Service by enabling:

- a higher number of active video cameras simultaneously,
- the transmission of high-resolution (4K) videos
- a better efficiency of communications between operators in the field and operations room thanks to the reduced latency.



UseCase – Static infrastructure monitoring IoT

Scenario

Static structural monitoring applicable to buildings and infrastructures. It's based on:

- Network of wireless sensors that collect and send data through an NB-IoT gateway
- Platform for data collection and historicization
- User application for viewing and analyzing collected data

It is possible to measure static parameters (eg cracks dilation, structure inclination) and correlate them to environmental parameters such as temperature and humidity



UseCase – Smart Industry - Advanced Maintenance

Scenario

Advanced Maintenance system for the management of the maintenance processes of equipment, devices and complex structures through advanced support tools, both in the logic of **On-Condition management** (with advanced **Prognostic and Predictive Maintenance** techniques), and through **Augmented Reality** features made available to **technical personnel**.

The system makes it possible to **predict failures**, thanks to the **correlation of data from remote sensors**, thus providing an indication of:

- **estimate of the residual life** of the equipment
- identification of the **most probable faults**

Acquisition & Storage

Equipment, Sensors, IoT DATA

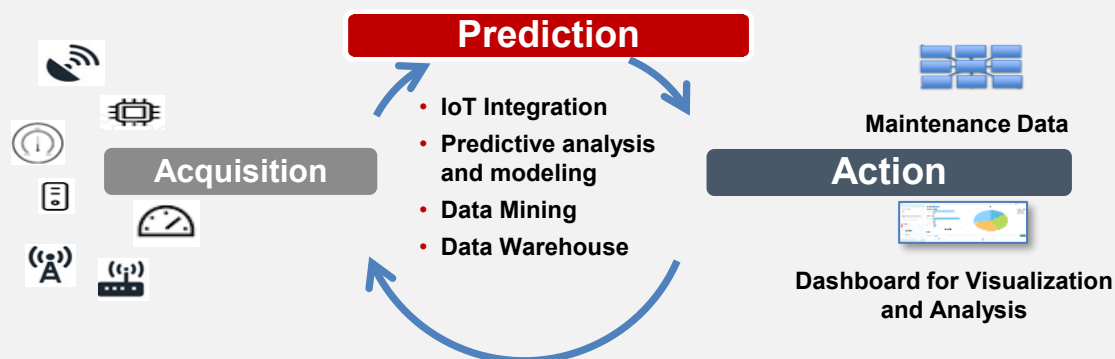
Vibrations,
Pressure,
....

Environmental DATA (e.g.,
Temperature, Humidity,,,...)

BUSINESS DATA

Mathematical data, technical data....

Real-Time Analysis and Processing



Real-Time Integration and Decision-Making

- **Health Monitoring**
- **Early warning**
- **Activation of Automatic Maintenance**
- **Design Improvements**
- **Root Cause Analysis and Problem Management**
- **Inventory Optimization and Spare Parts Management**

UseCase – Smart Industry – Industrial AGV

Scenario

Smart Manufacturing at the level of production area and neighboring environments (between **production and warehouses** or **between production cells**), replacing traditional **logistic systems** such as **conveyors and carts** with human guidance.

The **autonomous logistics** is realized through **self-guided vehicles (Automated Guided Vehicles - AGV)** managed from **control algorithms** in the IT systems.

Connection between the AGV and the control layer is supported **by 5G**.

AGV systems are often the “**retrofit**” of traditional vehicles already in use such as **forklifts, cranes, shuttles** and heavy machines.

Some **planned applications**:

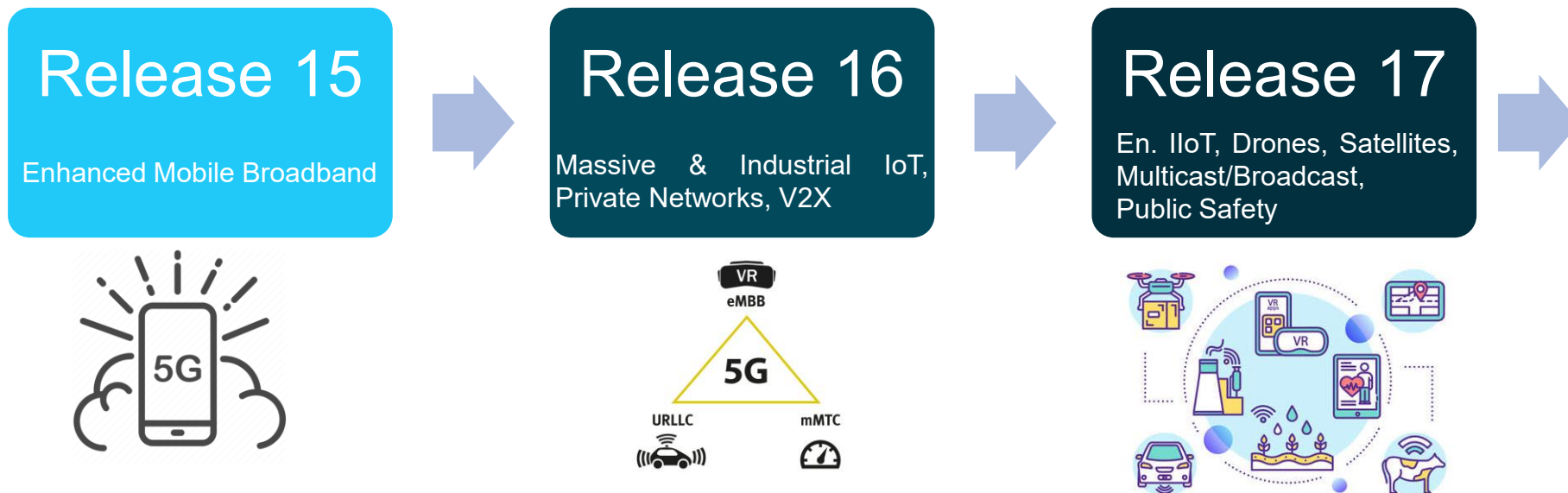
- AGVs operating inside the factory for **logistics between production cells**
- AGVs operating between factory environment for the **logistics between production and warehouses** in separate buildings or outdoor loading areas
- Use cases are envisaged in the **port area** with AGVs such as bridge cranes and cranes operating at the docks,
- **loading / unloading of goods and containers**, and at warehouses.



5G evolution towards 6G



5G Phases (1/2)



5G Phase 1 (June 2019)

- Enhanced Mobile Broadband (**eMBB**)
- Basic URLLC
- Deployment options (opt.3, 2 and 5, 4 and 7)

5G Phase 2 (July 2020)

- Capacity enhancements
- Operational efficiency
- Expansion to **new vertical markets**

5G expansion (June 2022)

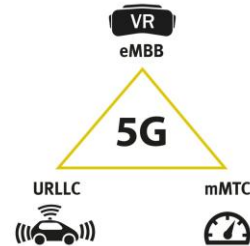
- Even **more vertical markets**
- Continuous enhancements

5G Phases (2/2)



5G Advanced (June 2024)

- AI/ML
- RedCap
- Extended Reality (XR)
- Network Energy Efficiency



5G Adv Phase 2 (Sept 2025)

- MIMO/D-MIMO improvements
- Faster mobility management
- Upgrade to XR/NTN/RedCap
- Sensing and AI/ML



5G expansion (March 2027?)

- Finalize 5G-Adv
- First studies on 6G (ISAC, advanced NTN, native AI, sustainability, ...)

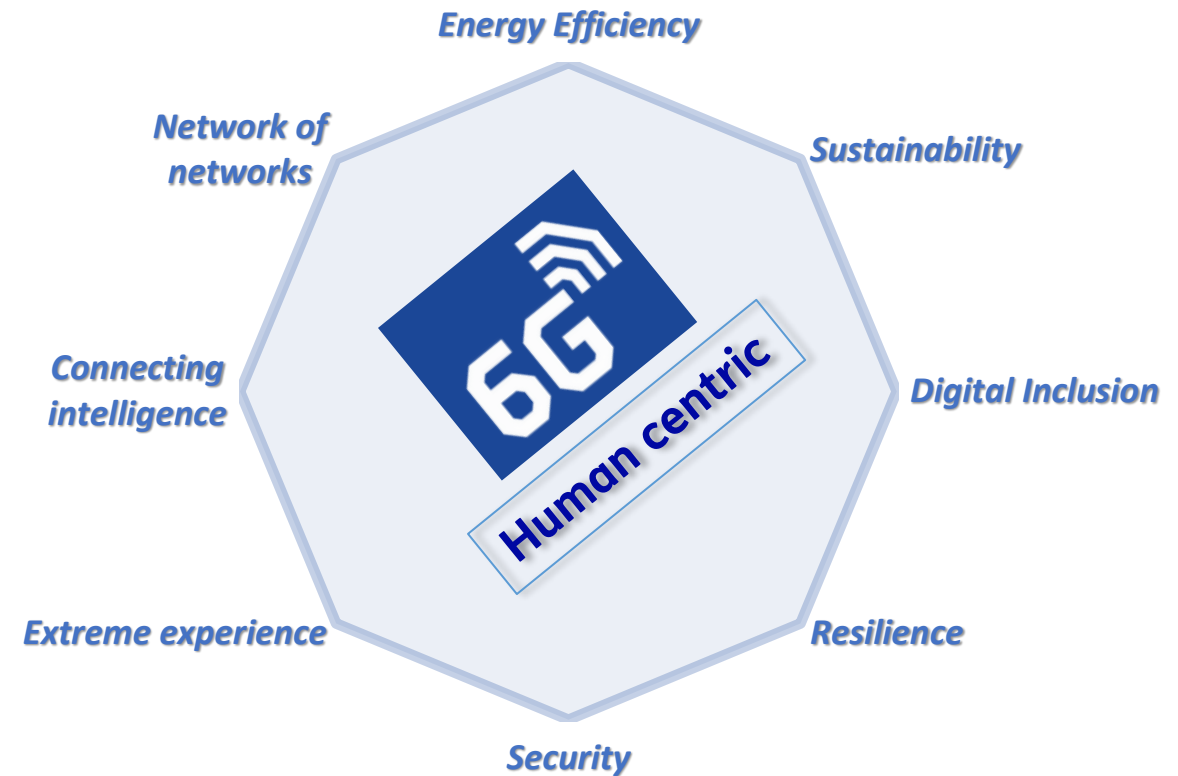
TIM View on evolutionary trends

6G should not be yet another radio technology providing “10x” performance: it should rather respond to future society needs

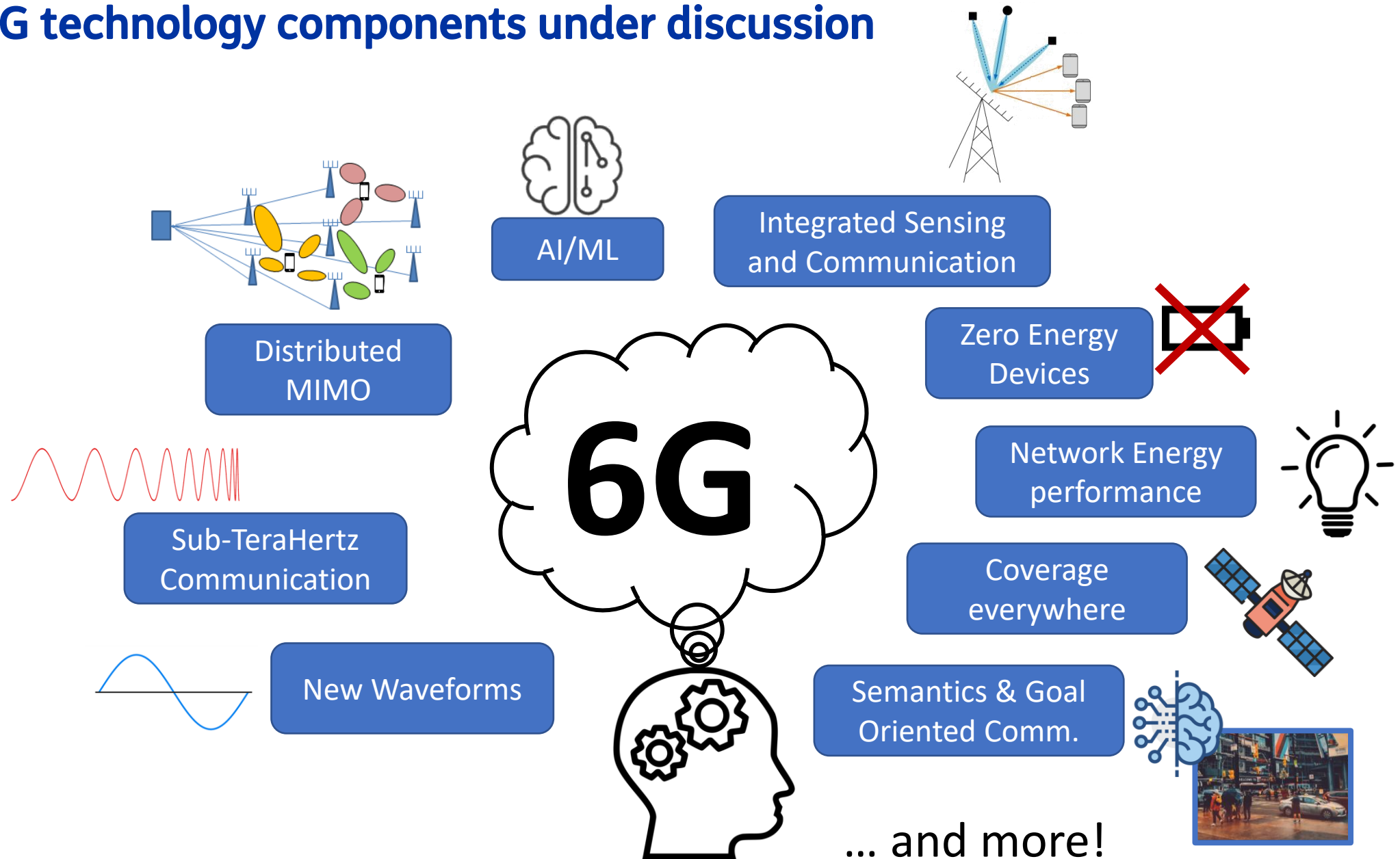
Interdisciplinary approach

International cooperation towards research
and standardization

Timeframe: **around 2030**



6G technology components under discussion



GRAZIE